

April 15th 2025

Graphene synthesis project update for previous week

Objective 1: Larger scale production of biochar (rotating oven)

MB3063

Objective: Material production at 1:1.5 ratio and 50 g batch size

Experiment details

50.0 g biochar (pilot plant batch #1, KFT 3.6%) milled (Blendtec, 10 sec) with 0.7 g water, then milled (Blendtec, 3x30 sec) with 75.0 g KOH, unloaded in glove box -> 119.8 g light brown powder, no color change

Rotating oven, 800 °C, 1 h, 3 °C/min, 119.4 g powder used, 22.8 g (130 ml) output

→ temperature ramp slightly different due to defective thermoelement

Result: Normal yield (48%), Species 1 (combined sample MB3063/MB3064/MB3066/MRa420)

MB3064

Objective: Material production at 1:1.5 ratio and 50 g batch size

Experiment details

50.0 g biochar (20.2 g pilot plant batch #1, KFT 3.6% + 29.8 g pilot plant batch #2, KFT 4.3%) milled (Blendtec, 10 sec) with 0.6 g water, then milled (Blendtec, 3x30 sec) with 75.0 g KOH, unloaded in glove box -> 118.7 g light brown powder, no color change

Rotating oven, 800 °C, 1 h, 3 °C/min, 118.0 g powder used, 21.4 g (120 ml) output

→ temperature ramp slightly different due to defective thermoelement

Result: Normal yield (45%), Species 1 (combined sample MB3063/MB3064/MB3066/MRa420)

MB3065

Objective: Material production at 1:1.5 ratio and 50 g batch size

Experiment details

50.0 g biochar (pilot plant batch #2, KFT 4.3%) milled (Blendtec, 10 sec) with 0.4 g water, then milled (Blendtec, 3x30 sec) with 75.0 g KOH, unloaded in glove box -> 123.2 g light brown powder, no color change

Rotating oven, 800 °C, 1 h, 3 °C/min, 122.4 g powder used, 22.3 g (120 ml) output

→ temperature ramp slightly different due to defective thermoelement

→ material removed from tube with (drinking) water jet

Result: Normal yield (46%), (mostly) species 1, comparable to previous experiments

Conclusion: Standard production batch, no apparent material spillover, drinking water workup appears feasible

Recommended action: Discussion on adaption of workup protocol

MB3066

Objective: Material production at 1:1.5 ratio and 50 g batch size

Experiment details

50.0 g biochar (pilot plant batch #2, KFT 4.3%) milled (Blendtec, 10 sec) with 0.4 g water, then milled (Blendtec, 3x30 sec) with 75.0 g KOH, unloaded in glove box -> 123.7 g light brown powder, no color change

Rotating oven, 800 °C, 1 h, 3 °C/min, 123.1 g powder used, 23.8 g (130 ml) output

→ temperature ramp slightly different due to defective thermoelement

Result: Normal yield (48%), Species 1 (combined sample MB3063/MB3064/MB3066/MRa420)

MRa420

Objective: Material production at 1:1.5 ratio and 50 g batch size

Experiment details

50.1 g biochar (pilot plant batch #2, KFT 4.3%) milled (Blendtec, 10 sec) with 0.4 g water, then milled (Blendtec, 3x30 sec) with 75.0 g KOH, unloaded in glove box -> 123.8 g light brown powder, no color change

Rotating oven, 800 °C, 1 h, 3 °C/min, 123.2 g powder used, 23.3 g (135 ml) output

→ temperature ramp slightly different due to defective thermoelement

Result: Normal yield (47%), Species 1 (combined sample MB3063/MB3064/MB3066/MRa420)

MRa423

Objective: Material production at 1:1.5 ratio and 50 g batch size

Experiment details

50.0 g biochar (pilot plant batch #2, KFT 4.3%) milled (Blendtec, 10 sec) with 0.4 g water, then milled (Blendtec, 3x30 sec) with 75.0 g KOH, unloaded in glove box -> 123.7 g light brown powder, no color change

Rotating oven, 800 °C, 1 h, 3 °C/min, 123.5 g powder used, 23.5 g (130 ml) output

→ temperature ramp slightly different due to defective thermoelement

Result: Normal yield (48%), SEM results pending

MB3067

Objective: Material production at 1:1.5 ratio and 50 g batch size

Experiment details

50.0 g biochar (pilot plant batch #2, KFT 4.3%) milled (Blendtec, 10 sec) with 0.4 g water, then milled (Blendtec, 3x30 sec) with 75.0 g KOH, unloaded in glove box -> 117.4 g light brown powder, no color change

Rotating oven, 800 °C, 1 h, 3 °C/min, 116.6 g powder used, 21.9 g (125 ml) output

→ temperature ramp slightly different due to defective thermoelement

Result: Normal yield (47%), SEM results pending

MB3068

Objective: Material production at 1:1.5 ratio and 50 g batch size

Experiment details

50.0 g biochar (pilot plant batch #2, KFT 4.3%) milled (Blendtec, 10 sec) with 0.4 g water, then milled (Blendtec, 3x30 sec) with 75.0 g KOH, unloaded in glove box -> 114.3 g light brown powder, no color change

Rotating oven, 800 °C, 1 h, 3 °C/min, 114.0 g powder used, 21.4 g (120 ml) output

→ temperature ramp slightly different due to defective thermoelement

Result: Normal yield (47%), SEM results pending

Overall status

- 35 rotating oven batches completed, total 450 g (mostly pure species 1) and 84 g (less pure, mixtures, species 2) product
- max batch size apparently reached or exceeded at 55-60 g biochar
- performance of 1:1 experiments on large scale unsatisfactory
- product quality consistency moderate, blender performance and biochar lot variability suspected cause for quality and consistency deficits
- product density variable (8-23 ml/g), down to 5 ml/g for 1:1 ratio
- current frequency is four batches (approx. 90 g product) per week, further increase in batch size not advisable
- step 1 production on lab scale stopped, 2/3 pilot plant batches finished, batch 3 in progress (material in drying)
- So far, most products only characterized by SEM, MRa389A and C also by conductivity measurement (results below)
- Production of species 2 feasible on 10-15 g batch size
- Revised workup (removal of material with water jet) tested and implemented, deionized water is used routinely, drinking water was tested

Objective 2: Testing/Optimization on small scale

MRa416/417/418

Objective: A series of side-by-side identical small-scale 1:1.5 tests with lab biochar as investigation in consistency

Experiment details

4.0 g biochar (various combined lots, KFT 4.7%) milled (3 min) in glovebox with 6.0 g KOH → 15x 0.6 g portions of brown powder, no change in appearance during bottling in glovebox

MRa416A: 800 °C, 1 h, 3 °C/min, 2 portions (2 x 0.6 g) pressed to pellets, standard inertion with nitrogen bottle (oven A), 234 mg (1 ml, 49% yield) output

→ Nitrogen stream increased over night (5 → 23.5), pressure regulator defective?

MRa416B: 800 °C, 1 h, 3 °C/min, 2 portions (2 x 0.6 g) pressed to pellets, standard inertion (oven B), 222 mg (1 ml, 46% yield) output

MRa417A: 800 °C, 1 h, 3 °C/min, 2 portions (2 x 0.6 g) pressed to pellets, standard inertion with nitrogen bottle (oven A), 189 mg (1 ml, 39% yield) output

→ Nitrogen stream increased over night (5 → >24), pressure regulator defective?

MRa417B: 800 °C, 1 h, 3 °C/min, 2 portions (2 x 0.6 g) pressed to pellets, standard inertion (oven B), 201 mg (1.7 ml, 42% yield) output

MRa418A: 800 °C, 1 h, 3 °C/min, 2 portions (2 x 0.6 g) pressed to pellets, standard inertion (oven A), 167 mg (1 ml, 35% yield) output

MRa418B: 800 °C, 1 h, 3 °C/min, 2 portions (2 x 0.6 g) pressed to pellets, standard inertion (oven B), 182 mg (1 ml, 38% yield) output

Result: (Mostly) species 1, notably higher ratio of foreign structures in MRa417B

Conclusion: Consistency improved, output (more) similar to expectations

Recommended action: Further investigation of consistency

MRa419/421

Objective: Identical small-scale 1:1.5 tests with pilot plant biochar as investigation in consistency

Experiment details

3.0 g biochar (pilot plant batch #2, KFT 4.3%) milled (1 min) with 0.024 g water, then milled (3 min) with 4.5 g KOH -> 8x 0.6 g portions of brown powder, no change in appearance during bottling

MRa419A: 800 °C, 1 h, 3 °C/min, 2 portions (2 x 0.6 g) pressed to pellets, standard inertion (oven A), 113 mg (2.7 ml, 26% yield) output

MRa419B: 800 °C, 1 h, 3 °C/min, 2 portions (2 x 0.6 g) pressed to pellets, standard inertion (oven B), 116 mg (2.7 ml, 24% yield) output

MRa421A: 800 °C, 1 h, 3 °C/min, 2 portions (2 x 0.6 g) pressed to pellets, standard inertion (oven A), 103 mg (2.8 ml, 23% yield) output

MRa421B: 800 °C, 1 h, 3 °C/min, 2 portions (2 x 0.6 g) pressed to pellets, standard inertion (oven B), 114 mg (3.2 ml, 26% yield) output

Result: Low yields, products rich in species 2, species ratio inconsistent

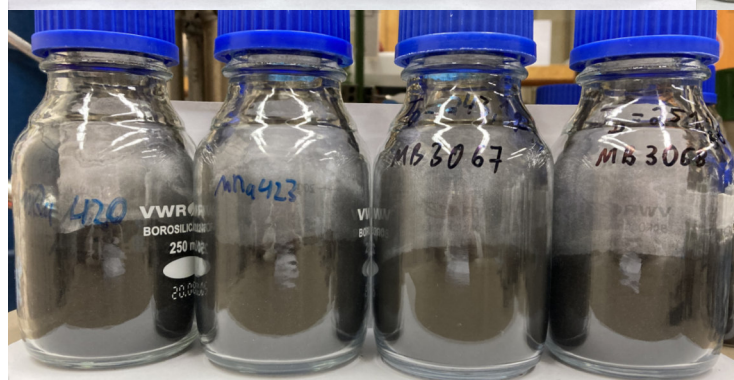
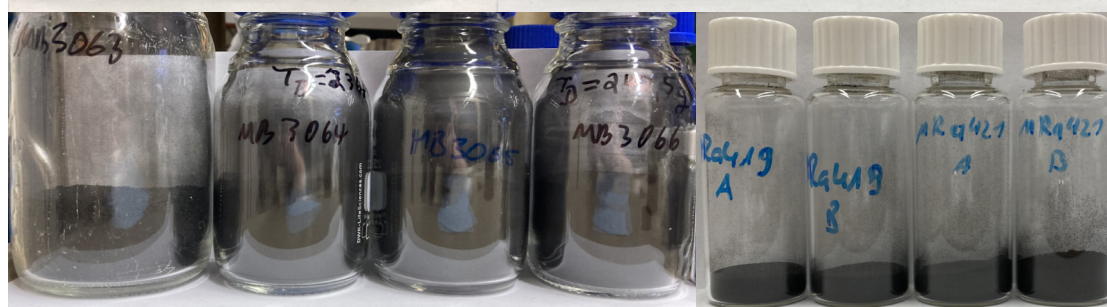
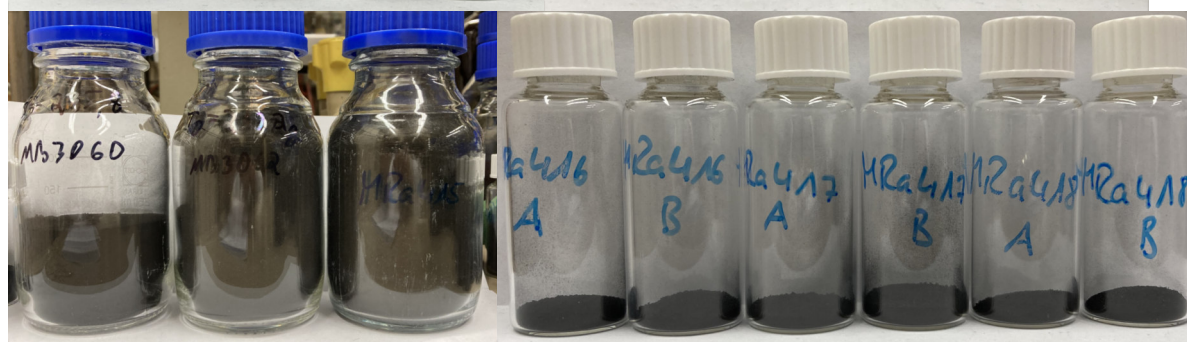
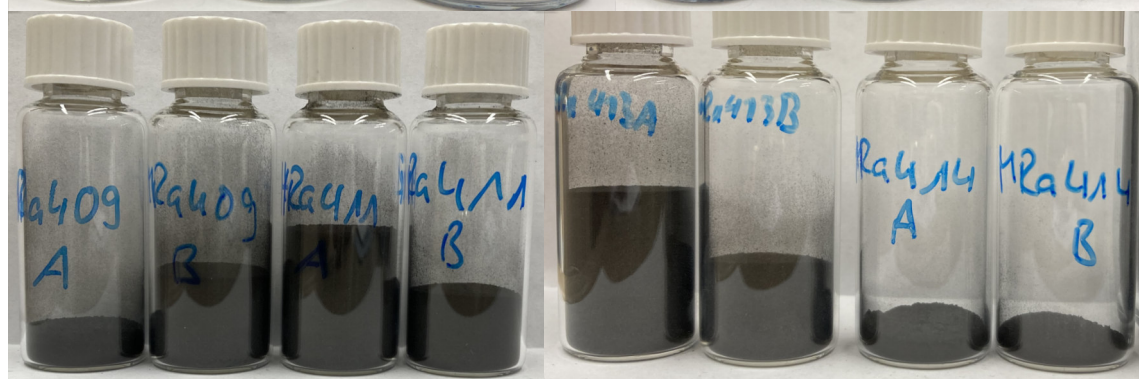
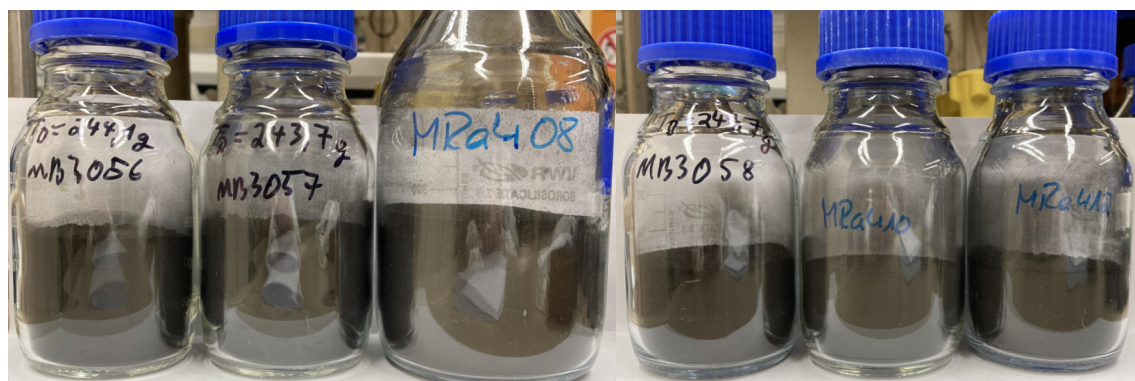
Conclusion: Consistency poor, selectivity for species 2 is very noteworthy deviation from previous set of experiments, likely causes are increased contact with air, different properties of biochar (different lot) or both

Recommended action: None

Overall status

- Latest changes (increased water content, 1:1 ratio biochar/KOH) not suitable or beneficial
- Cause of low consistency of outcomes currently not clear





Analytical data

- SEM (and partially Raman) results available on sharepoint for
 - TB1133, MB2946A, MB2947D, TB1135-1, TB1135-2, MB2952B, MB2952C, MB2952C2, MB2955A, MB2955A2, TB1136, MB2962A/B, MB2963A/B, TB1137-1/2/3 (step 2 products), MB2965B, MB2965B2, MB2966A/B, MB2967A/B, MB2970A/B, MB2971A/B, MB2972A/B, MB2973A/B, MB2974A/B, MB2975A/B, MB2976A/B, MB2979A/B, MB2980A/B/C/D, KJo-0165A/B, KJo-0166A/B, KJo-0167A/B, KJo-0168A/B, KJo-0169, KJo-0170, KJo-0171A/B, KJo-0172A/B, KJo-0175, KJo-0176, KJo-0177, KJo-0178, KJo-0171A GR, MB2981A, TB1138, MB2984, MRa320, TB1139, TB1140A, TB1140B, MB2988, MRa323, TB1141, MRa326, MRa327, MRa328, MB2990A, MB2990B, MRa329, MRa331A, MRa331B, MRa333A, MRa333B, MRa334A, MRa334B, MRa2993A, MRa2993B, MB2995A, MB2995B, MB2997A, MB2997B, TB1144A, MB1144B, MB2999A, MB2999B, TB1146, TB1147, MB3001A, MB3001B, MB3003A, MB3003B, MB3004A, MB3004B, TB1151A, TB1151B, MB3007A, MB3007B, MB3009, MB3013A, MB3013B, MB3015A, MB3015B, MRa340A, MRa340B, MB3017A, MB3017B, MB3019A, MB3019B, MB3021A, MB3021B, MB3023A, MB3023B, MRa352A, MRa352B, MRa353A, MRa353B, TB1158A, TB1158B, TB1142, TB1160A, TB1160B, TB1162A, TB1162B, MB3026, MB3027, MB3028, MB3030, TB1165, TB1166, MB3036, MB3038, MRa373, TB1167, MRa375, MB3041, MRa376, MB3043, MRa380, MRa385, MRa386, MB3046, MRa389A, MRa389B, MRa389C, MB3050, MB3051, MB3054, MRa394, MRa395, MRa398A, MRa398B, MB3055, MRa399, MRa401, MRa402, MRa403, MRa404, MRa405, MRa406, MRa407, MRa408, MB3056, MB3057, MRa409A, MRa409B, MRa410, MRa411A, MRa411B, MRa412, MB3058, MB3060, MRa414A, MRa414B, MB3062, MRa415, MRa416A, MRa416B, MRa417A, MRa417B, MRa418A, MRa418B, MB3065, MRa419A, MRa419B, MRa421A, MRa421B, MB3063/MB3064/MB3066/MRa420 (step 2 products)
 - 900561-500MG, 2 lots (Sigma-Aldrich reference)
 - 924458-500MG (Sigma-Aldrich reference “Single-layer graphene sheets for battery, Bio-sourced”)
 - Graphene oxide
- TEM results available for
 - MB2955A2 (step 2 product)
 - MB2976A (step 2 product)
 - TB1137-1 (step 2 product)

- BET results

Sample	Multipoint BET Area [m ² /g]	Langmuir Surface Area [m ² /g]	Species
MB2976A	1.88 x 10 ³	2.55 x 10 ³	mostly 1
MRa329	1.76 x 10 ³	2.39 x 10 ³	2
MRa333A	2.09 x 10 ³	5.74 x 10 ³	2
MRa334A	1.86 x 10 ³	2.50 x 10 ³	mostly 2
MB3001A	1.613 x 10 ³	7.763 x 10 ³	mostly 2
MB3004A	1.491 x 10 ³	2.032 x 10 ³	mostly 2
MB3001A/3004A	1.471 x 10 ³	1.983 x 10 ³	mostly 2
924458-500MG	1.112 x 10 ³	1.421 x 10 ³	n.a.
900561-500MG	1.942 x 10 ¹	2.442 x 10 ¹	n.a.
TB1142	1.257 x 10 ³	1.684 x 10 ³	1
TB1160B	1.240 x 10 ³	1.677 x 10 ³	1
TB1165	1.520 x 10 ³	2.020 x 10 ³	1

- Conductivity results

Material	Species	Data file	Conductivity (S/cm)				Comment
			1 kN	8 kN	12 kN	20 kN	
Species 1 (combined batches, "Coffee grinder"/pellet press/quartz tube oven)	1	TB1142-1	1.4	4.5	5.6	7.2	As sent to Maxwell Technologies and Dr Chris
		TB1142-2	1.4	4.6	5.7	7.4	
Mostly species 2 ("Coffee grinder"/pellet press/quartz tube oven) Lot MB3001A/3004A	Mostly 2	350645-1	5.0	14.5	17.5	22.5	
		350645-2	5.0	13.4	16.1	20.2	
"Single-layer graphene sheets for battery, Bio-sourced, avg. no. of layers, 1" according to Sigma-Aldrich	n.a.	924458-500MG-1	0.1	0.4	0.5	0.7	
		924458-500MG-2	0.1	0.4	0.5	0.7	
"Graphene, powder, electrical conductivity >10 ³ S/m, avg. no. of layers, < 3" according to Sigma-Aldrich	n.a.	900561-500MG-1	206.6	767.0	862.3	963.5	
		900561-500MG-2	204.7	768.5	863.8	976.6	
Norit Supac	n.a.	TB1159A-1	2.5	8.0	9.7	12.5	sample sent by Dr Li November 2024
		TB1159A-2	2.5	8.0	9.8	12.5	
Graphite	n.a.	TB1159B-1	51.3	205.8	256.2	329.3	sample sent by Dr Li November 2024
		TB1159B-2	51.0	201.4	251.6	322.7	
Species 1 (Blender/Rotating oven) Lot MRa389A	1	MRa389-A1	4.2	12.1	14.6	18.4	
		MRa389-A2	3.6	10.8	13.3	16.9	
Species 1 (Blender/Rotating oven) Lot MRa389C	1	MRa389-C1	4.1	11.9	14.5	18.4	
		MRa389-C2	4.2	12.2	14.8	18.8	
Species 2 (Blender/Rotating oven) Lot MRa408	2	MRa408-1	4.6	14.3	17.5	22.3	
		MRa408-2	4.6	14.1	17.2	21.9	

Experiment overview – Step 1

Experiment	Reactor	Raw material			Acid			T	t	pressure		Wash		Output		
MB2928	AV1	6.0 g	BAFA neu Hemp Fibre VF	100.0 g	0.19%	0.02 M	Sulfuric acid	180 °C	22.5 h	8.7 bar	-> 10.4 bar	742 g	Water	2.6 g		
Comment	Fibres not completely covered by liquid															
MB2933	AV1	6.1 g	BAFA neu Hemp Fibre VF	174.0 g	0.19%	0.02 M	Sulfuric acid	180 °C	24 h	8.7 bar	-> 9.5 bar	647 g	Water	1.2 g		
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MB2935	AV1	6.0 g	BAFA neu Hemp Fibre VF	172.9 g	0.19%	0.02 M	Sulfuric acid	180 °C	23 h	8.8 bar	-> 9.5 bar	500 g	Water	1.3 g		
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MB2936	AV1	6.0 g	BAFA neu Hemp Fibre VF	172.4 g	0.96%	0.10 M	Sulfuric acid	180 °C	23 h	8.8 bar	-> 10.6 bar	589 g	Water	0.8 g	Combined and homogenized as lot no MRa231	
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MRa225	AV1	6.0 g	BAFA neu Hemp Fibre VF	153.8 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	?	-> 10.5 bar	500 g	Water	0.9 g		
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MRa228	AV1	6.3 g	BAFA neu Hemp Fibre VF	154.0 g	0.96%	0.10 M	Sulfuric acid	180 °C	22.5 h	8.7 bar	-> 11.0 bar	500 g	Water	0.9 g		
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MRa231	AV1	6.1 g	BAFA neu Hemp Fibre VF	168.8 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	8.0 bar	-> 10.5 bar	500 g	Water	0.8 g		
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MB2937	AV1	9.4 g	Canadian Rockies Hemp	179.3 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.6 bar	-> 10.9 bar	569 g	Water	1.6 g	Combined and homogenized as lot no MB2946X	
Comment	very low amount of foreign material observed and removed during filtration															
MB2938	AV1	8.9 g	Canadian Rockies Hemp	175.0 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	9.2 bar	-> 13.2 bar	577 g	Water	1.4 g		
Comment	very low amount of foreign material observed and removed during filtration															
MB2940	AV1	9.2 g	Canadian Rockies Hemp	177.1 g	0.96%	0.10 M	Sulfuric acid	180 °C	22.5 h	8.7 bar	-> 10.6 bar	555 g	Water	1.5 g		
Comment	very low amount of foreign material observed and removed during filtration															
MB2942	AV1	8.9 g	Canadian Rockies Hemp	177.6 g	0.96%	0.10 M	Sulfuric acid	180 °C	23 h	8.3 bar	-> 13.5 bar	578 g	Water	1.6 g		
Comment	low amount of foreign material observed and removed during filtration															
MB2943	AV1	9.8 g	Canadian Rockies Hemp	178.2 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.3 bar	-> 12.7 bar	611 g	Water	1.7 g		
Comment	low amount of foreign material observed and removed during filtration															
MB2944	AV1	9.0 g	Canadian Rockies Hemp	178.4 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.4 bar	-> 9.7 bar	593 g	Water	1.8 g		
Comment	low amount of foreign material observed and removed during filtration															
MB2945	AV1	9.1 g	Canadian Rockies Hemp	177.7 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.7 bar	-> 9.8 bar	6 g	Water	1.6 g		
Comment	low amount of foreign material observed and removed during filtration															
MB2948	AV1	9.4 g	Canadian Rockies Hemp	175.1 g	0.96%	0.10 M	Sulfuric acid	180 °C	25 h	9.0 bar	-> 12.6 bar	475 g	Water	1.7 g		
Comment	low amount of foreign material observed and removed during filtration															

Experiment	Reactor	Raw material		Acid			T	t	pressure		Wash		Output	Drying	KFT
MB2949	AV1	9.2 g	Canadian Rockies Hemp	174.3 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	8.4 bar -> 10.2 bar	538 g	Water	1.8 g		
Comment	low amount of foreign material observed and removed during filtration														
MB2950	AV1	9.1 g	Canadian Rockies Hemp	170.5 g	0.96%	0.10 M	Sulfuric acid	180 °C	23 h	-> 11.1 bar	534 g	Water	1.6 g		
Comment	low amount of foreign material observed and removed during filtration														
MB2951	AV1	9.3 g	Canadian Rockies Hemp	178.5 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	9.1 bar -> 11.2 bar	575 g	Water	1.8 g		
Comment	low amount of foreign material observed and removed during filtration														
MRa255	AV1	9.2 g	Canadian Rockies Hemp	170.8 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	8.1 bar -> 12.8 bar	488 g	Water	1.7 g		
Comment	some foreign material observed and removed during filtration														
MB2953	AV1	9.7 g	Canadian Rockies Hemp	177.1 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	9.1 bar -> 10.4 bar	558 g	Water	1.8 g		
Comment	low amount of foreign material observed and removed during filtration														
MB2956	AV1	9.8 g	Canadian Rockies Hemp	172.0 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.6 bar -> 12.5 bar	548 g	Water	1.9 g		
Comment	low amount of foreign material observed and removed during filtration														
MB2957	AV1	9.1 g	Canadian Rockies Hemp	176.8 g	1.50%	0.15 M	Sulfuric acid	180 °C	23.5 h	8.8 bar -> 8.5 bar	589 g	Water	1.5 g		
Comment															
MB2958	AV1	9.3 g	Canadian Rockies Hemp	177.0 g	2.00%	0.20 M	Sulfuric acid	180 °C	24 h	8.9 bar -> 8.7 bar	562 g	Water	1.8 g		
Comment															
MB2959	AV5	76.0 g	Canadian Rockies Hemp	953.0 g	1.00%	0.10 M	Sulfuric acid	180 °C	25 h		3957 g	Water	14.9 g		
Comment	low amount of foreign material observed and removed during filtration														
MB2960	AV1	3.8 g	Canadian Rockies Hemp (cut)	151.0 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h	8.8 bar -> 10.5 bar	300 g	Water	0.6 g		
Comment	Reaction performed with stirring														
MB2961	AV1	9.9 g	Canadian Rockies Hemp	172.0 g	5.00%	0.51 M	Sulfuric acid	180 °C	24 h	10.1 bar -> 16.1 bar	563 g	Water	1.7 g		
Comment	low amount of foreign material observed and removed during filtration														
MB2964	AV1	9.0 g	Canadian Rockies Hemp	211.0 g + 140.8 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h	8.6 bar -> 8.8 bar	310 g + 603 g	Water	1.3 g		
Comment	Hemp boiled with 1% H2SO4 at 95 °C, filtered and washed, then heated in Autoclave with fresh acid														
MB2978	AV5	76.3 g	Canadian Rockies Hemp	822.0 g	1.00%	0.10 M	Sulfuric acid	180 °C	22 h		4112 g	Water	15.4 g		
Comment															
KJo-0173	AV5	77.1 g	Canadian Rockies Hemp	942.1 g	1.00%	0.10 M	Sulfuric acid	185 °C	24 h		4283 g	Water	16.5 g	40 °C	4.9%
Comment														100 °C	2.3%
KJo-0174	AV5	77.1 g	Canadian Rockies Hemp	945.0 g	1.00%	0.10 M	Sulfuric acid	190 °C	24 h		4384 g	Water	15.8 g	40 °C	3.7%
Comment															
MB2982	AV5	77.1 g	Canadian Rockies Hemp	999.5 g	1.00%	0.10 M	Sulfuric acid	190 °C	24 h		4258 g	Water	14.9 g	40 °C	5.3%
Comment															
MB2983	AV5	80.5 g	Canadian Rockies Hemp	999.8 g	1.00%	0.10 M	Sulfuric acid	190 °C	24 h		4232 g	Water	16.6 g	40 °C	5.5%
Comment														100 °C	1.7%

Experiment	Reactor	Raw material	Acid	T	t	pressure	Wash	Output	Drying	KFT
MB2985	AV5	79.5 g Canadian Rockies Hemp	999.7 g 1.00% 0.10 M Sulfuric acid	200 °C	24 h		4241 g Water	15.4 g	40 °C	3.7%
Comment									100 °C	1.9%
MB2986	AV5	79.9 g Canadian Rockies Hemp	999.9 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4128 g Water	16.0 g	60 °C	2.7%
Comment									100 °C	1.7%
MB2989	AV5	80.2 g Canadian Rockies Hemp	999.3 g 1.00% 0.10 M Sulfuric acid	180 °C	23 h		4314 g Water	18.2 g	40 °C	5.9%
Comment										
MB2991	AV5	81.3 g Canadian Rockies Hemp	999.3 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4223 g Water	19.1 g	40 °C	5.7%
Comment										
MB2992	AV5	81.5 g Canadian Rockies Hemp	999.7 g 1.00% 0.10 M Sulfuric acid	160 °C	23.5 h		4262 g Water	19.5 g	40 °C	6.2%
Comment										
MB2994	AV5	80.8 g Canadian Rockies Hemp	999.8 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4159 g Water	20.2 g	40 °C	5.2%
Comment										
MB2996	AV5	82.7 g Canadian Rockies Hemp	999.7 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4210 g Water	20.0 g	40 °C	4.6%
Comment										
MB2998	AV5	84.0 g Canadian Rockies Hemp	999.6 g 1.00% 0.10 M Sulfuric acid	180 °C	22 h		4263 g Water	19.8 g	40 °C	5.0%
Comment	Included nickel and steel material samples for corrosion/cleaning tests									
MB3000	AV5	82.5 g Canadian Rockies Hemp	1000.0 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4431 g Water	19.8 g	40 °C	5.0%
Comment										
MB3002	AV5	80.0 g Canadian Rockies Hemp	999.9 g 1.00% 0.10 M Sulfuric acid	190 °C	25 h		4245 g Water	18.5 g	40 °C	2.0%
Comment										
MB3005	AV5	81.4 g Canadian Rockies Hemp	999.4 g 1.00% 0.10 M Sulfuric acid	200 °C	24 h		4280 g Water	19.0 g	40 °C	2.0%
Comment										
MB3006	AV5	80.8 g Canadian Rockies Hemp	1000.1 g 1.00% 0.10 M Sulfuric acid	190 °C	23 h		4182 g Water	17.4 g	40 °C	2.1%
Comment										
MB3008	AV5	82.2 g Canadian Rockies Hemp	1000.2 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4238 g Water	18.0 g	40 °C	2.6%
Comment										
MB3010	AV5	81.3 g Canadian Rockies Hemp	1000.0 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4235 g Water	18.9 g	40 °C	2.0%
Comment										
MB3011	AV5	82.4 g Canadian Rockies Hemp	999.4 g 1.00% 0.10 M Sulfuric acid	160 °C	24 h		4244 g Water	23.5 g	40 °C	2.2%
Comment										
MB3014	AV5	82.8 g Canadian Rockies Hemp	999.8 g 1.00% 0.10 M Sulfuric acid	160 °C	23 h		1460 g Water	23.6 g	40 °C	1.3%
Comment										
MRa341	AV5	82.3 g Canadian Rockies Hemp	1000.4 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		1232 g Water	19.2 g	40 °C	2.1%
Comment										

Experiment	Reactor	Raw material	Acid	T	t	pressure	Wash	Output	Drying	KFT
MB3016	AV5	81.7 g Canadian Rockies Hemp	999.7 g 1.00% 0.10 M Sulfuric acid	185 °C	21 h		4232 g Water	19.7 g	40 °C	2.1%
Comment										
MB3018	AV5	81.2 g Canadian Rockies Hemp	1000.1 g 1.00% 0.10 M Sulfuric acid	195 °C	22 h		4266 g Water	19.0 g	40 °C	2.1%
Comment										
MB3020	AV5	81.1 g Canadian Rockies Hemp	1000.2 g 1.00% 0.10 M Sulfuric acid	185 °C	26 h		4278 g Water	18.9 g	40 °C	1.7%
Comment										
MB3022	AV5	81.8 g Canadian Rockies Hemp	999.9 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4262 g Water	19.0 g	40 °C	2.9%
Comment										
MB3024	AV5	80.7 g Canadian Rockies Hemp	999.6 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4256 g Water	18.4 g	40 °C	2.0%
Comment										
MRa349	AV5	80.8 g Canadian Rockies Hemp	999.4 g 1.00% 0.10 M Sulfuric acid	185 °C	26 h		4254 g Water	19.1 g	40 °C	2.2%
Comment										
MRa354	AV5	80.2 g Canadian Rockies Hemp	999.7 g 1.00% 0.10 M Sulfuric acid	185 °C	26 h		4414 g Water	18.4 g	40 °C	1.7%
Comment										
MRa355	AV5	80.2 g Canadian Rockies Hemp	999.5 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		4262 g Water	18.2 g	40 °C	1.8%
Comment										
MRa358	AV5	80.8 g Canadian Rockies Hemp	999.9 g 1.00% 0.10 M Sulfuric acid	180 °C	24 h		3317 g Water	18.8 g	40 °C	2.1%
Comment										
MB3025	AV5	81.4 g Canadian Rockies Hemp	1000.5 g 1.00% 0.10 M Sulfuric acid	165 °C	21.5 h		4229 g Water	18.4 g	40 °C	1.1%
Comment										
MB3029	AV5	81.6 g Canadian Rockies Hemp	999.7 g 1.00% 0.10 M Sulfuric acid	185 °C	26 h		4178 g Water	18.8 g	40 °C	1.7%
Comment										
MB3031	AV5	82.5 g Canadian Rockies Hemp	999.7 g 1.00% 0.10 M Sulfuric acid	190 °C	25 h		4178 g Water	19.8 g	40 °C	1.5%
Comment										
MB3032	AV5	81.9 g Canadian Rockies Hemp	1000.5 g 1.00% 0.10 M Sulfuric acid	180 °C	23.5 h		4341 g Water	19.0 g	rt	4.8%
Comment										
MRa367	AV5	81.9 g Canadian Rockies Hemp	1000.0 g 1.00% 0.10 M Sulfuric acid	190 °C	24.5 h		4256 g Water	19.8 g	rt	4.7%
Comment										
MRa368	AV5	81.4 g Canadian Rockies Hemp	999.8 g 1.00% 0.10 M Sulfuric acid	185 °C	27.5 h		4290 g Water	20.3 g	rt	5.5%
Comment										
MB3034	AV5	84.7 g Canadian Rockies Hemp	1000.2 g 1.00% 0.10 M Sulfuric acid	180 °C	24.5 h		4254 g Water	20.0 g	40 °C	2.0%
Comment										
MB3037	AV5	80.7 g Canadian Rockies Hemp	999.8 g 1.00% 0.10 M Sulfuric acid	185 °C	21.5 h		4325 g Water	18.6 g	40 °C	1.7%
Comment										

Experiment	Reactor	Raw material	Acid	T	t	pressure	Wash	Output	Drying	KFT
MB3039	AV5	81.2 g Canadian Rockies Hemp	1000.0 g 1.00% 0.10 M Sulfuric acid	180 °C	25 h		4253 g Water	19.1 g	40 °C	1.8%
Comment										
MB3040	AV5	80.0 g Canadian Rockies Hemp	1002.2 g 1.00% 0.10 M Sulfuric acid	185 °C	26 h		4301 g Water	18.8 g	40 °C	
Comment									rt	4.3%
MRa3042	AV5	80.0 g Canadian Rockies Hemp	999.9 g 1.00% 0.10 M Sulfuric acid	185 °C	26 h		4026 g Water	19.3 g	40 °C	
Comment									rt	4.3%
MB3044	AV5	82.4 g Canadian Rockies Hemp	1000.0 g 1.00% 0.10 M Sulfuric acid	180 °C	25 h		4316 g Water	20.2 g	40 °C	
Comment									rt	4.5%
MB3045	AV5	82.8 g Canadian Rockies Hemp	1000.1 g 1.00% 0.10 M Sulfuric acid	160 °C	23 h		3964 g Water	22.8 g	rt	5.1%
Comment										
MB3047	AV5	81.1 g Canadian Rockies Hemp	1000.1 g 1.00% 0.10 M Sulfuric acid	160 °C	24 h		3836 g Water	22.7 g	rt	5.3%
Comment										
MB3048	AV5	80.2 g Canadian Rockies Hemp	1000.2 g 1.00% 0.10 M Sulfuric acid	160 °C	23 h		4275 g Water	23.9 g	rt	3.6%
Comment										
MB3049	AV5	80.7 g Canadian Rockies Hemp	999.7 g 1.00% 0.10 M Sulfuric acid	155 °C	25 h		4248 g Water	23.3 g	rt	3.6%
Comment										

Comments

- Red text for T and t shows experiments where the temperature tracks have not yet been evaluated (or are not available)

Experiment overview – Step 2

Experiment	Oven	Qty	Lot	Base	grinding	Gas	T (rate)	T (max)	t	Wash	Drying	Output
TB1131	A	1.2 g	MB2933	1.2 g KOH (90%)	manual	Ar	20-27 °C/min	790 °C	1 h	20 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.16 g
Comment		ground biochar: brown powder, strong electrostatic charge										
TB1132	A	1.3 g	MB2935	1.3 g KOH (90%)	mill (2 min)	Ar	20-29 °C/min	792 °C	1 h	20 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.3 g
Comment		ground biochar: brown powder										
TB1133	A	1.0 g	MRa231	1.0 g KOH (90%)	mill (1 min)	Ar	3 °C/min	785 °C	1 h	20 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.3 g
Comment		ground biochar: brown powder; see temperature trend										
TB1134	A	1.0 g	MRa231	1.0 g KOH (90%)	mill (2 min)	Ar	3 °C/min	771 °C	1 h	20 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.3 g
Comment		ground biochar: brown powder; see temperature trend										
MB2946	A	1.0 g	MB2946X	1.0 g KOH (90%)	mill (1 min)	Ar	5 °C/min	770 °C	1 h	22 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.3 g
Comment		ground biochar: brown powder; see temperature trend										
MB2947A	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	Ar	3 °C/min	803 °C	1 h	10 g 10% HCl (+ water)	40 °C N2 stream 210 mbar	< 0.01 g
Comment		ground biochar: brown powder; see temperature trend										
MB2947B	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	Ar	3 °C/min	804 °C	2 h	10 g 10% HCl (+ water)	40 °C N2 stream 210 mbar	< 0.01 g
Comment		ground biochar: brown powder; see temperature trend										
MB2947C	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	Ar	3 °C/min	806 °C	3 h	Batch discarded		
Comment		ground biochar: brown powder; see temperature trend										
MB2947D	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	Ar	3 °C/min	810 °C	3 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	38 mg
Comment		ground biochar: brown powder; see temperature trend										
TB1135-1	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	14 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.06 g
Comment		ground biochar: brown powder; see temperature trend										
TB1135-2	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	14 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.07 g
Comment		ground biochar: brown powder; see temperature trend										
MB2952B	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.07 g
Comment		ground biochar: brown powder; see temperature trend										
MB2952C	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	3 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.05 g
Comment		ground biochar: brown powder; see temperature trend										

Experiment	Oven	Qty	Lot	Base	grinding	Gas	T (rate)	T (max)	t	Wash	Drying	Output
MB2952C2	A	0.25 g	MB2946X	0.25 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2955A	A	0.25 g	MB2946X	0.38 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	7 mg
Comment		Quartz tube shattered during experiment								(+ water)	(atm. Pressure)	
MB2955A2	A	0.25 g	MB2946X	0.38 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.09 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
TB1136	A	0.26 g	MB2946X	0.39 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	14 g 10% HCl	100 °C N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2962A	A	0.25 g	MB2946X	0.50 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2962B	A	0.25 g	MB2946X	0.50 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.09 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2963A	A	0.22 g	MB2960	0.22 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2963B	A	0.29 g	MB2960	0.29 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.12 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
TB1137-1	A	0.26 g	MB2946X	1.06 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	23 g 10% HCl	100 °C N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
TB1137-2	A	0.28 g	MB2946X	1.13 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	23 g 10% HCl	100 °C N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
TB1137-3	A	0.26 g	MB2946X	1.06 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	23 g 10% HCl	100 °C N2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2965A	A	0.22 g	MB2959	1.08 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	20 g 10% HCl	100 °C N2 stream	0.02 g
Comment		ground biochar/KOH mixture likely absorbed some moisture; see temperature trend								(+ water)	(atm. Pressure)	
MB2965B	A	0.22 g	MB2959	1.08 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	20 g 10% HCl	100 °C N2 stream	0.01 g
Comment		ground biochar/KOH mixture likely absorbed some moisture; see temperature trend								(+ water)	(atm. Pressure)	
MB2965B2	B	0.22 g	MB2959	1.08 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	20 g 10% HCl	100 °C N2 stream	0.01 g
Comment		ground biochar/KOH mixture likely absorbed some moisture; see temperature trend								(+ water)	(atm. Pressure)	

Experiment	Oven	Qty	Lot	Base	grinding	Gas	T (rate)	T (max)	t	Wash	Drying	Output
MB2966A	B	0.26 g	MB2959	1.04 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	20 g 10% HCl	100 °C N2 stream	0.05 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2966B	A	0.26 g	MB2959	1.04 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	20 g 10% HCl	100 °C N2 stream	0.05 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2967A	B	0.26 g	MB2964	1.04 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	20 g 10% HCl	100 °C N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2967B	A	0.26 g	MB2964	1.04 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	20 g 10% HCl	100 °C N2 stream	0.04 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2970A	B	0.24 g	MB2957	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2970B	A	0.24 g	MB2957	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.02 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2971A	B	0.24 g	MB2958	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.05 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2971B	A	0.24 g	MB2958	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2972A	B	0.24 g	MB2961	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.09 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2972B	A	0.24 g	MB2961	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2973A	B	0.24 g	MB2959	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.04 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2973B	A	0.24 g	MB2959	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.09 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2974A	B	0.24 g	MB2959	0.36 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2974B	A	0.24 g	MB2959	0.36 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	

Experiment	Oven	Qty	Lot	Base	grinding	Gas	T (rate)	T (max)	t	Wash	Drying	Output
MB2975A	B	0.24 g	MB2959	0.36 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2975B	A	0.24 g	MB2959	0.36 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2976A	B	0.24 g	MB2959	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.08 g
Comment		ground biochar (brown powder) compacted to pellet; see temperature trend								(+ water)	(atm. Pressure)	
MB2976B	A	0.24 g	MB2959	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl	100 °C N2 stream	0.11 g
Comment		ground biochar (brown powder) compacted to pellet; see temperature trend								(+ water)	(atm. Pressure)	
MB2979A	B	0.24 g	MB2959	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2979B	A	0.24 g	MB2959	0.36 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.10 g
Comment		ground biochar: brown powder; see temperature trend								(+ water)	(atm. Pressure)	
MB2980A	B	0.24 g	MB2959/2956	0.36 g KOH (90%)	mill (20 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.08 g
		ground biochar (brown powder) compacted to pellet								(+ water)	(atm. Pressure)	
MB2980B	A	0.24 g	MB2959/2956	0.36 g KOH (90%)	mill (10 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.10 g
		ground biochar (brown powder) compacted to pellet								(+ water)	(atm. Pressure)	
MB2980C	B	0.52 g	MB2959/2956	0.78 g KOH (90%)	mill (20 min)	N2	3 °C/min	800 °C	1 h	21 g 10% HCl	100 °C N2 stream	0.16 g
		ground biochar (brown powder) compacted to pellet								(+ water)	(atm. Pressure)	
MB2980D	A	0.52 g	MB2959/2956	0.78 g KOH (90%)	mill (10 min)	N2	3 °C/min	800 °C	1 h	21 g 10% HCl	100 °C N2 stream	0.23 g
		ground biochar (brown powder) compacted to pellet								(+ water)	(atm. Pressure)	
KJo-0165A	A	0.24 g	MB2978	0.36 g KOH (90%)	mill (10 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl	100 °C N2 stream	0.09 g
		ground biochar: brown powder								(+ water)	(atm. Pressure)	
KJo-0165B	B	0.24 g	MB2978	0.36 g KOH (90%)	mill (20 min)	N2	3 °C/min	800 °C	1 h	11 g 10% HCl	100 °C N2 stream	0.15 g
		ground biochar: brown powder								(+ water)	(atm. Pressure)	
KJo-0166A	A	0.24 g	MB2978	0.36 g KOH (90%)	mill (10 min)	N2	10 °C/min	800 °C	1 h	11 g 10% HCl	100 °C N2 stream	0.13 g
		ground biochar: brown powder								(+ water)	(atm. Pressure)	
KJo-0166B	B	0.24 g	MB2978	0.36 g KOH (90%)	mill (20 min)	N2	10 °C/min	800 °C	1 h	11 g 10% HCl	100 °C N2 stream	0.11 g
		ground biochar: brown powder								(+ water)	(atm. Pressure)	

Experiment	Oven	Qty	Lot	Base	grinding	Gas	T (rate)	T (max)	t	Wash	Drying	Output
KJo-0167A	A	0.24 g	MB2978	0.36 g KOH (90%)	mill (10 min)	N2	3 °C/min	800 °C	4 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.10 g
		ground biochar (brown powder) compacted to pellet										
KJo-0167B	B	0.24 g	MB2978	0.36 g KOH (90%)	mill (20 min)	N2	3 °C/min	800 °C	4 h	11 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.06 g
		ground biochar (brown powder) compacted to pellet										
KJo-0168A	A	0.24 g	MB2978	0.36 g KOH (90%)	mill (10 min)	N2	3 °C/min	900 °C	2 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.02 g
		ground biochar (brown powder) compacted to pellet										
KJo-0168B	B	0.24 g	MB2978	0.36 g KOH (90%)	mill (20 min)	N2	3 °C/min	900 °C	1 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.09 g
		ground biochar (brown powder) compacted to pellet										
KJo-0169	B	0.80 g	MB2978	1.20 g KOH (90%)	mill (10 min)	N2	3 °C/min	800 °C	1 h	30 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.23 g
		ground biochar (brown powder) compacted to pellet										
KJo-0170	B	1.07 g	MB2978	1.61 g KOH (90%)	mill (20 min)	N2	3 °C/min	800 °C	1 h	38 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.62 g
		ground biochar (brown powder) compacted to pellet										
KJo-0171A	B	1.60 g	MB2978	2.40 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	60 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.64 g
		ground biochar (brown powder) compacted to two pellets of equal size										
KJo-0171B	A	0.24 g	MB2978	0.36 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.13 g
		ground biochar (brown powder) compacted to pellet										
KJo-0172A	B	0.50 g	MB2978	0.50 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	11 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.24 g
		ground biochar (brown powder) compacted to pellet										
KJo-0172B	A	0.50 g	MB2978	0.50 g KOH (90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.17 g
		ground biochar (brown powder) compacted to pellet										
KJo-0175	A	1.60 g	MB2978	2.40 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	61 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.71 g
		ground biochar (brown powder) compacted to two pellets of equal size, quartz tube shattered										
KJo-0176	B	0.24 g	MB2978	0.36 g KOH (90%)	mill (5 min)	N2	1.5 °C/min	800 °C	1 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.10 g
		ground biochar (brown powder) compacted to pellet										
KJo-0177	A	1.60 g	MB2978	2.40 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	61 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.91 g
		ground biochar (brown powder) compacted to two pellets of equal size										
KJo-0178	A	1.60 g	MB2955/74/75/76	2.40 g KOH (90%)	mill (1-5 min)	N2	3 °C/min	800 °C	1 h	60 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.55 g
		ground biochar (brown powder) compacted to two pellets of equal size										
MB2981	B	1.44 g	MB2955/74/75/76	2.16 g KOH (90%)	mill (1-5 min)	N2	3 °C/min	800 °C	1 h	60 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.44 g
		ground biochar (brown powder) compacted to six pellets of equal size										

Experiment	Oven	Qty	Lot	Base	grinding	homogeneous?	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Volume	Species	Appearance
TB1138	A	1.54 g	various	2.31 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	59 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.59 g 38%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa320	B	1.43 g	various	2.14 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	53 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.55 g 38%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2984	B	1.40 g	various	2.10 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	53 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.48 g 34%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1139	A	1.39 g	KJo-0173	2.08 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	50 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.54 g 39%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1140A	A	1.28 g	KJo-0173	1.92 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	49 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.46 g 36%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1140B	B	1.28 g	KJo-0173	1.92 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	50 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.48 g 38%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa323	A	1.06 g	KJo-0173	1.60 g KOH (90%)	mill (3 min)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.44 g 41%		1	black/grey brittle
		ground biochar (sonicated, brown powder) compacted to two pellets of equal size														
TB1141	A	1.08 g	various	1.62 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	30 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.40 g 37%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa326	A	1.28 g	KJo-0173	1.92 g KOH (90%)	mill (3 min)		N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.51 g 39%		1	black/grey brittle
		ground biochar (sonicated, brown powder) compacted to two pellets of equal size														
MRa327	B	1.04 g	KJo-0173	1.56 g KOH (90%)	mill (3 min)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.46 g 44%		1	black/grey brittle
		ground biochar (sonicated, brown powder) compacted to two pellets of equal size														
MB2988	B	1.04 g	KJo-0173	1.56 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	45 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.42 g 40%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa328	B	1.20 g	KJo-0173	1.80 g KOH (90%)	mill (3 min)		N2	3 °C/min	800 °C	1 h	45 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.53 g 44%		1	black/grey brittle
		ground biochar (sonicated, brown powder) compacted to two pellets of equal size														
MRa329	A	1.04 g	KJo-0173 (extra dried)	1.56 g KOH (90%)	mill (1 min)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.43 g 42%		2	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2990A	A	1.28 g	MB2982	1.92 g KOH (90%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.50 g 39%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2990B	B	1.04 g	MB2982	1.56 g KOH (90%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.42 g 40%		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														

Experiment	Oven	Qty	Lot	Base	grinding	homogeneous?	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Vol/Dens	Species	Appearance
MRa331A	B	1.28 g	MB2985 (extra dried)	1.92 g KOH (90%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.64 g 50%		Mostly 1	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa331B	A	1.04 g	MB2985 (extra dried)	1.56 g KOH (90%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.46 g 44%		Mostly 1	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa333A	A	1.28 g	MB2985 (extra dried)	1.15 g KOH (90%) 0.77 g NaOH (98%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.21 g 17%		2	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa333B	B	1.04 g	MB2985 (extra dried)	0.94 g KOH (90%) 0.62 g NaOH (98%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.26 g 25%		2	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa334A	A	1.04 g	MB2985 (extra dried)	1.25 g KOH (90%) 0.31 g NaOH (98%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.48 g 46%		Mostly 2	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa334B	B	0.88 g	MB2985 (extra dried)	1.06 g KOH (90%) 0.26 g NaOH (98%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	33 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.39 g 45%		Mostly 2	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2993A	A	1.28 g	MB2982	1.15 g KOH (90%) 0.77 g NaOH (98%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.22 g 18%		Mostly 2	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2993B	B	1.04 g	MB2982	0.94 g KOH (90%) 0.62 g NaOH (98%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.19 g 18%		Mostly 2	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2995A	A	1.28 g	MB2983 (extra dried)	1.92 g KOH (90%)	mill (4 min)		N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.54 g 42%	3.9 ml 7.3 ml/g	Mostly 1	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2995B	B	1.04 g	MB2983 (extra dried)	1.56 g KOH (90%)	mill (4 min)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.46 g 44%	5.6 ml 12.3 ml/g	1/2 mix	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1144A	A	1.04 g	KJo-0174	1.56 g KOH (90%)	ball mill (15 Hz, 80 sec)	no	N2	3 °C/min	800 °C	1 h	38 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.61 g 59%	5.0 ml 8.2 ml/g	1 + fibres	black/grey somewhat shiny
		ground biochar (brown powder) NOT compacted														
TB1144B	B	1.28 g	KJo-0174	1.92 g KOH (90%)	ball mill (15 Hz, 80 sec)	no	N2	3 °C/min	800 °C	1 h	54 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.36 g 28%	6.0 ml 16.9 ml/g	2 (?)	black/grey shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2997A	A	1.28 g	MB2985 (extra dried)	1.92 g KOH (90%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.55 g 43%	3.6 ml 6.6 ml/g	1	black/grey somewhat shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2997B	B	1.04 g	MB2985 (extra dried)	1.56 g KOH (90%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	40 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.45 g 43%	2.9 ml 6.4 ml/g	1	black/grey somewhat shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
MB2999A	A	1.28 g	KJo-0174	1.92 g KOH (90%)	ball mill (25 Hz, 80 sec)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.68 g 53%	4.6 ml 6.8 ml/g	Mostly 1/2 mix	shiny black/grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														

Experiment	Oven	Qty	Lot	Base	grinding	homogeneous?	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Vol/Dens	Species	Appearance
MB2999B	B	1.04 g	KJo-0174	1.56 g KOH (90%)	ball mill (25 Hz, 80 sec)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.40 g 38%	4.6 ml 11.5 ml/g	1 + fibres	shiny black/grey voluminous
		ground biochar (brown powder) NOT compacted														
TB1146	A	1.28 g	MB2982	1.92 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	51 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.52 g 41%	4.0 ml 7.6 ml/g	1	black/grey somewhat shiny
		ground biochar/KOH mix (brown powder, MB2990) dried at 100 °C, then compacted to two pellets of equal size														
TB1147	B	1.08 g	MB2983 (extra dried)	1.62 g KOH (90%)	mill (4 min)		N2	3 °C/min	800 °C	1 h	37 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.36 g 33%	5.0 ml 14.1 ml/g	Mostly 1	black/grey somewhat shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
MB3001A	A	1.28 g	KJo-0174	1.92 g KOH (90%)	ball mill (35 Hz, 80 sec)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.49 g 38%	10.6 ml 21.5 ml/g	Mostly 2	shiny black/grey very voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB3001B	B	1.04 g	KJo-0174	1.56 g KOH (90%)	ball mill (35 Hz, 80 sec)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.48 g 46%	8.2 ml 17.2 ml/g	Mostly 2	shiny black/grey very voluminous
		ground biochar (brown powder) NOT compacted														
MB3003A	A	1.28 g	MB2986 (2% water)	1.92 g KOH (90%)	ball mill (35 Hz, 80 sec)		N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.57 g 45%	5.7 ml 9.9 ml/g	Mostly 1	black/grey somewhat shiny
		freshly ground biochar (brown powder) compacted to two pellets of equal size														
MB3003B	B	1.04 g	MB2986 (2% water)	1.56 g KOH (90%)	ball mill (35 Hz, 80 sec)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.44 g 42%	9.0 ml 20.7 ml/g	Mostly 1/2 mix	shiny black/grey voluminous
		freshly ground biochar (brown powder) NOT compacted														
MB3004A	A	1.28 g	MB2986 (2% water)	1.92 g KOH (90%)	ball mill (35 Hz, 80 sec)		N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.59 g 46%	17.4 ml 29.4 ml/g	Mostly 2	shiny black/grey very voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB3004B	B	1.04 g	MB2986 (2% water)	1.56 g KOH (90%)	ball mill (35 Hz, 80 sec)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.41 g 39%	8.4 ml 20.7 ml/g	Mostly 1/2 mix	shiny black/grey voluminous
		ground biochar (brown powder) NOT compacted														
TB1151A	A	1.40 g	KJo-0174	2.10 g KOH (90%)	ball mill (15 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	51 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.53 g 38%	2.5 ml 4.7 ml/g	1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1151B	B	1.08 g	KJo-0174	1.62 g KOH (90%)	ball mill (15 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.45 g 42%	2.5 ml 5.6 ml/g	1	black/grey brittle
		ground biochar (brown powder) NOT compacted														
MB3007A	A	1.20 g	KJo-0174	1.80 g KOH (90%)	ball mill (15 Hz, 30 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.50 g 42%	3.0 ml 6.0 ml/g	1	black/grey barely shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
MB3007B	B	1.04 g	KJo-0174	1.56 g KOH (90%)	ball mill (15 Hz, 30 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.43 g 42%	2.9 ml 6.7 ml/g	1	black/grey barely shiny
		ground biochar (brown powder) NOT compacted														
MB3009	A	1.28 g	MB2992	1.92 g KOH (90%)	mill (1 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.41 g 32%	3.0 ml 7.4 ml/g	1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size														
MB3012A	A	1.28 g	MB2983 (2% water)	1.92 g KOH (90%)	ball mill (35 Hz, 10+1 min)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.64 g 50%	10.2 ml 16.0 ml/g	1/2 mix	shiny black/grey voluminous
		ground biochar (brown powder) NOT compacted														

Experiment	Oven	Qty	Lot		Base		grinding		homogeneous?	Gas	T (rate)	T (max)	t	Wash		Drying		Output	Vol/Dens	Species	Appearance
MB3012B	B	1.04 g	MB2983 (2% water)	1.56 g	KOH (90%)			ball mill (15 Hz, 10+1 min)	no	N2	3 °C/min	800 °C	1 h	39 g	10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.44 g 42%	10.8 ml 24.6 ml/g	Mostly 1/2 mix	shiny black/grey voluminous
		ground biochar (brown powder) NOT compacted																			
MB3013A	A	1.28 g	MB2983 (2% water)	1.92 g	KOH (90%)			ball mill (35 Hz, 10+1 min)	no	N2	3 °C/min	800 °C	1 h	48 g	10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.58 g 45%	4.3 ml 7.5 ml/g	Mostly 1/2 mix	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size																			
MB3013B	B	1.04 g	MB2983 (2% water)	1.56 g	KOH (90%)			ball mill (15 Hz, 10+1 min)	no	N2	3 °C/min	800 °C	1 h	39 g	10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.40 g 38%	12.8 ml 32.2 ml/g	Mostly 2	shiny black/grey very voluminous
		ground biochar (brown powder) compacted to two pellets of equal size																			
MB3015A	A	1.20 g	MB2986 (2% water)	1.80 g	KOH (90%)			ball mill (15 Hz, 30+1 min)	no	N2	3 °C/min	800 °C	1 h	45 g	10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.39 g 33%	8.0 ml 20.5 ml/g	Mostly 2	shiny black/grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size																			
MB3015B	B	1.00 g	MB2986 (2% water)	1.50 g	KOH (90%)			ball mill (15 Hz, 30+1 min)	no	N2	3 °C/min	800 °C	1 h	39 g	10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.36 g 36%	8.0 ml 22.5 ml/g	Mostly 2	shiny black/grey voluminous
		ground biochar (brown powder) NOT compacted																			
MRa340A	A	1.28 g	MB2996	1.92 g	KOH (90%)			ball mill (15 Hz, 120 min)	yes	N2	3 °C/min	800 °C	1 h	48 g	10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.49 g 38%	4.0 ml 8.2 ml/g	1	black/grey barely shiny
		ground biochar (brown powder) compacted to two pellets of equal size																			
MRa340B	B	1.04 g	MB2996	1.56 g	KOH (90%)			ball mill (15 Hz, 120 min)	yes	N2	3 °C/min	800 °C	1 h	39 g	10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.44 g 42%	3.5 ml 7.9 ml/g	1	black/grey barely shiny
		ground biochar (brown powder) NOT compacted																			
TB1155A	A	1.28 g	MB2986 (2% water)	1.54 g	KOH (90%)	0.38 g	NaOH (98%)	ball mill (15 Hz, 10 min)	no	N2	3 °C/min	800 °C	1 h	48 g	10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.69 g 54%	4.0 ml 5.8 ml/g		black/grey somewhat shiny
		NaOH not fully ground!, ground biochar (brown powder) compacted to two pellets of equal size																			
TB1155B	B	1.04 g	MB2986 (2% water)	1.25 g	KOH (90%)	0.31 g	NaOH (98%)	ball mill (15 Hz, 10 min)	no	N2	3 °C/min	800 °C	1 h	39 g	10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.45 g 43%	4.0 ml 8.9 ml/g		black/grey somewhat shiny
		NaOH not fully ground!, ground biochar (brown powder) NOT compacted																			
MB3017A	A	1.28 g	MB3002 (2% water)	1.54 g	KOH (90%)	0.38 g	NaOH (98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	48 g	10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.53 g 41%	19.1 ml 36.3 ml/g	2	shiny black/grey very voluminous
		ground biochar (brown powder) compacted to two pellets of equal size																			
MB3017B	B	1.04 g	MB3002 (2% water)	1.25 g	KOH (90%)	0.31 g	NaOH (98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	39 g	10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.33 g 32%	12.2 ml 36.7 ml/g	2	shiny black/grey voluminous
		ground biochar (brown powder) NOT compacted																			
MB3019A	A	1.28 g	MB3002 (2% water)	1.92 g	KOH (90%)			ball mill (15 Hz, 10+1 min)	no	N2	3 °C/min	800 °C	1 h	48 g	10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.47 g 37%	15.9 ml 33.9 ml/g	2	shiny black/grey very voluminous
		ground biochar (brown powder) compacted to two pellets of equal size																			
MB3019B	B	1.04 g	MB3002 (2% water)	1.56 g	KOH (90%)			ball mill (15 Hz, 10+1 min)	no	N2	3 °C/min	800 °C	1 h	39 g	10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.39 g 38%	9.3 ml 23.7 ml/g	Mostly 2	shiny black/grey voluminous
		ground biochar (brown powder) NOT compacted																			
MB3021A	A	1.28 g	MB3002 (2% water)	1.92 g	KOH (90%)			mill (2 min)		N2	3 °C/min	800 °C	1 h	48 g	10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.52 g 41%	7.3 ml 14.0 ml/g	Mostly 1	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size																			
MB3021B	B	1.04 g	MB3002 (2% water)	1.56 g	KOH (90%)			mill (2 min)		N2	3 °C/min	800 °C	1 h	39 g	10% HCl (+ water)	100 °C	N2 stream (atm. Pressure)	0.43 g 42%	5.3 ml 12.2 ml/g	Mostly 1	shiny black/grey
		ground biochar (brown powder) NOT compacted																			

Experiment	Oven	Qty	Lot	Base	grinding	homogeneous?	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Vol/Dens	Species	Appearance
MB3023A	A	1.08 g	MB2989	1.30 g KOH (90%) 0.32 g NaOH (98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.28 g 26%	16.5 ml 59.1 ml/g	2	shiny black/grey very voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB3023B	B	0.80 g	MB2989	0.96 g KOH (90%) 0.24 g NaOH (98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.25 g 31%	13.5 ml 54.7 ml/g	Mostly 2	shiny black/grey very voluminous
		ground biochar (brown powder) NOT compacted														
MRa352A	A	1.28 g	MB3002 (2% water)	1.92 g KOH (90%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.62 g 48%	2.9 ml 4.7 ml/g	1	black/grey barely shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa352B	B	1.04 g	MB3002 (2% water)	1.56 g KOH (90%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.53 g 51%	2.8 ml 5.3 ml/g	1	black/grey barely shiny
		ground biochar (brown powder) NOT compacted														
MB353A	A	1.28 g	MB2991	1.92 g KOH (90%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.53 g 41%	2.0 ml 3.8 ml/g	1	black/grey barely shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa353B	B	1.04 g	MB2991	1.39 g KOH (90%) 0.17 g NaOH (98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.44 g 43%	3.5 ml 7.9 ml/g	Mostly 1	black/grey barely shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1158A	A	1.04 g	MB2991	1.25 g KOH (90%) 0.31 g NaOH (98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	50 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.40 g 39%	17.0 ml 42.3 ml/g	2	shiny black/grey very voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1158B	B	0.96 g	MB2991	1.37 g KOH (90%) 0.08 g NaOH (98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	45 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.39 g 40%	3.0 ml 7.8 ml/g	Mostly 1	black/grey barely shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1160A	A	1.04 g	MB2991	1.25 g KOH (90%) 0.31 g NaOH (98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.46 g 44%	17.0 ml 37.0 ml/g	2	shiny black/grey very voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1160B	B	1.04 g	MB2991	1.56 g KOH (90%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	41 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.43 g 41%	2.0 ml 4.6 ml/g	1	black/grey barely shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1162A	A	1.04 g	MB2989	1.56 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	45 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.44 g 43%	4.0 ml 9.0 ml/g	Mostly 1	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1162B	B	1.04 g	MB2989	1.25 g KOH (90%) 0.31 g NaOH (98%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	45 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.32 g 31%	8.0 ml 25.2 ml/g	Mostly 2	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size														
MB3026	B	1.04 g	MB3005 (2% water)	1.56 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.40 g 39%	8.3 ml 20.6 ml/g	1/2 mix	shiny black/grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB3027	B	1.04 g	MB3005 (2% water)	1.25 g KOH (90%) 0.31 g NaOH (98%)	mill (3 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.28 g 27%	18.0 ml 64.7 ml/g	2	shiny black/grey very voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB3028	B	1.04 g	MB2989	1.56 g KOH (90%)	mill (10 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.42 g 40%	5.1 ml 12.1 ml/g	Mostly 1	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size														

Experiment	Oven	Qty	Lot	Base	grinding	homogeneous?	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Vol/Dens	Species	Appearance
MB3030	A	1.04 g	MB2989	1.25 g KOH (90%) 0.31 g NaOH (98%)	mill (10 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.29 g 28%	16.8 ml 57.7 ml/g	2	shiny black/grey very voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
TB1165	C	12.4 g	MB2992	18.5 g KOH (90%)	5 millings (3 min each)	yes	N2	3 °C/min	800 °C	1 h	193 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	4.55 g 37%	35.0 ml 7.7 ml/g	1	dull black/grey
		Rotating oven, powder not compacted														
TB1166	C	7.8 g	MB3006 (2% water)	9.41 g KOH (90%)	4 millings (3 min each)	yes	N2	3 °C/min	800 °C	1 h	210 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	3.25 g 41%	65.0 ml 20.0 ml/g	1/2 mix	shiny black/grey
		Rotating oven, powder not compacted														
MB3036	B	1.04 g	MB2992 + H2O	1.56 g KOH (90%)	mill (1+3 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.34 g 32%	3.0 ml 8.9 ml/g	1	black/grey barely shiny
		water added to mill => 7-8% (calc), ground biochar (brown powder) compacted to two pellets of equal size														
TB1167	C	14.8 g	MB2994	22.3 g KOH (90%)	blender, 90 sec	yes	N2	3 °C/min	800 °C	1 h	200 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	6.40 g 43%	50.0 ml 7.8 ml/g	Mostly 1	shiny black/grey
		Rotating oven, powder not compacted														
MB3038	B	1.04 g	MB2994	1.56 g KOH (90%)	blender, 90 sec	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.53 g 51%	2.4 ml 4.5 ml/g	Mostly 1	black/grey barely shiny
		ground biochar (brown powder from TB1167) compacted to two pellets of equal size														
MRa373	A	1.04 g	MB2989 + H2O	1.56 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.43 g 41%	3.5 ml 8.2 ml/g	Mostly 1	shiny black/grey
		water added to biochar => 8% (KFT), ground biochar (brown powder) compacted to two pellets of equal size														
MRa375	A	1.04 g	MB2996 + H2O	1.56 g KOH (90%)	mill (1+3 min)	yes	N2	3 °C/min	800 °C	1 h	40 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.51 g 49%	3.2 ml 6.3 ml/g	Mostly 1	black/grey barely shiny
		water added to mill => 9% (calc), ground biochar (brown powder) compacted to two pellets of equal size														
MRa376	C	16.7 g	MB2998 + H2O	25.0 g KOH (90%)	blender, 10+80 sec	?	N2	3 °C/min	800 °C	1 h	189 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	7.6 g 46%	60.0 ml 7.9 ml/g	Mostly 1	shiny black/grey
		water added to blender => 7-8% (calc), rotating oven, powder not compacted														
MB3041	B	1.3 g	MB2996	1.3 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.46 g 35%	3.0 ml 6.5 ml/g	1	black/grey barely shiny
		ground biochar (brown powder) compacted to two pellets of equal size														
MB3043	C	16.5 g	MB3000	24.72 g KOH (90%)	blender, 90 sec	yes	N2	3 °C/min	800 °C	1 h	285 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	7.11 g 43%	70.0 ml 9.8 ml/g	Mostly 1	black/grey barely shiny
		Rotating oven, powder not compacted														
MRa380	A	1.0 g	MB2996 + H2O	1.0 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.48 g 46%	5.0 ml 10.4 ml/g	1/2 mix	shiny black/grey
		water added to biochar => 11.5% (KFT), ground biochar (brown powder) compacted to two pellets of equal size														
MB3046	C	15.6 g	MB3032	24.20 g KOH (90%)	blender, 90 sec	yes	N2	3 °C/min	800 °C	1 h	311 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	7.05 g 45%	95.0 ml 13.5 ml/g	*	shiny black/grey
		KOH not milled, rotating oven, powder not compacted														
MRa385	C	17.8 g	MRa367	26.6 g KOH (90%)	blender, 6x15 sec	no	N2	3 °C/min	800 °C	1 h	183 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	8.60 g 48%	125.0 ml 14.5 ml/g	Mostly 1/2 mix	shiny black/grey
		Rotating oven, powder not compacted														
MRa386	A	2.1 g	MRa368	2.1 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	60 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.65 g 32%	3.7 ml 5.7 ml/g	1	black/grey barely shiny
		ground biochar (brown powder) compacted to two pellets of equal size														

Experiment	Oven	Qty	Lot	Base	grinding	homogeneous?	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Vol/Dens	Species	Appearance
MRa389A	C	17.2 g	MB3005/3008/3010 + H2O	25.8 g KOH (90%)	blender, 2x45 sec	?	N2	3 °C/min	800 °C	1 h	178 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	8.2 g 48%	70.0 ml 8.5 ml/g	1	black/grey barely shiny
			water added to blender => 5% (calc), rotating oven, powder not compacted													
MRa389B	C	16.5 g	MB3005/3008/3010 + H2O	24.8 g KOH (90%)	blender, 2x45 sec	?	N2	3 °C/min	800 °C	1 h	114 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	5.2 g 31%	50.0 ml 9.6 ml/g	1	black/grey somewhat shiny
			water added to blender => 5% (calc), rotating oven, powder not compacted													
MRa389C	C	15.2 g	MB3005/3008/3010 + H2O	22.9 g KOH (90%)	blender, 2x45 sec	?	N2	3 °C/min	800 °C	1 h	154 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	7.3 g 48%	60.0 ml	1	black/grey barely shiny
			water added to blender => 5% (calc), rotating oven, powder not compacted													
MB3050	C	17.8 g	MB3044	17.8 g KOH (90%)	blender, 90 sec	yes	N2	3 °C/min	800 °C	1 h	125 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	8.1 g 46%	40.0 ml 4.9 ml/g	Mostly 1*	dull black/grey
			Rotating oven, powder not compacted													
MB3051	C	16.3 g	MB3040	24.5 g KOH (90%)	blender, 4x45 sec	yes	N2	3 °C/min	800 °C	1 h	269 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	7.20 g 44%	90.0 ml 12.5 ml/g	Mostly 1	black/grey somewhat shiny
			Rotating oven, powder not compacted													
MRa394	C	21.4 g	MB3045	21.4 g KOH (90%)	blender, 90 sec	no	N2	3 °C/min	800 °C	1 h	148 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	7.2 g 34%	40.0 ml 5.5 ml/g	Mostly 1*	dull black/grey
			Rotating oven, powder not compacted													
MB3054	C	16.0 g	MB3034	16.0 g KOH (90%)	blender, 90 sec	no	N2	3 °C/min	800 °C	1 h	124 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	7.51 g 47%	70.0 ml 9.3 ml/g	*	dull black/grey
			Rotating oven, powder not compacted													
MRa395	A	1.3 g	MB3034	1.3 g KOH (90%)	blender, 90 sec	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.6 g 46%	3.0 ml 5.0 ml/g	Mostly 1*	dull black/grey
			ground biochar (brown powder) compacted to two pellets of equal size													
MRa398A	C	25.6 g	MB3016/3018/3020	38.5 g KOH (90%)	blender, 2x45 sec	no	N2	3 °C/min	800 °C	1 h	256 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	12.6 g 49%	100.0 ml 7.9 ml/g	1	dull black/grey
			Rotating oven, powder not compacted, powder turned dark during milling													
MRa398B	C	30.4 g	MB3016/3018/3020	45.6 g KOH (90%)	blender, 2x45 sec	no	N2	3 °C/min	800 °C	1 h	303 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	14.2 g 47%	90.0 ml 6.3 ml/g	1	dull black/grey
			Rotating oven, powder not compacted, powder turned dark during milling													
MB3055	C	40.6 g	(various combined lots)	61.0 g KOH (90%)	blender, 2x45 sec	no	N2	3 °C/min	800 °C	1 h	573 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	19.4 g 48%	105.0 ml 5.4 ml/g	1	dull black/grey
			Rotating oven, powder not compacted, powder turned dark during milling													
MRa399	C	47.7 g	(various combined lots)	71.5 g KOH (90%)	blender, 3x30 sec	no	N2	3 °C/min	800 °C	1 h	488 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	21.7 g 46%	125.0 ml 5.8 ml/g	1	dull black/grey
			Rotating oven, powder not compacted													
MRa401	C	48.7 g	TB1168 (pilot plant #1)	73.1 g KOH (90%)	blender, 3x30 sec	no	N2	3 °C/min	800 °C	1 h	517 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	23.9 g 49%	125.0 ml 5.2 ml/g	1	dull black/grey
			Rotating oven, powder not compacted													
MRa402	C	48.8 g	TB1168 (pilot plant #1) + H2O	73.3 g KOH (90%)	blender, 10 + 3x30 sec	no	N2	3 °C/min	800 °C	1 h	500 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	23.6 g 48%	110.0 ml 4.7 ml/g	1	dull black/grey
			Rotating oven, powder not compacted													
MRa403	C	44.9 g	(various combined lots)	67.4 g KOH (90%)	blender, 2x45 sec	no	N2	3 °C/min	800 °C	1 h	504 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	21.4 g 48%	120.0 ml 5.6 ml/g	1	dull black/grey
			Rotating oven, powder not compacted, powder turned dark during milling													

Experiment	Oven	Qty	Lot	Base	grinding	homogeneous?	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Vol/Dens	Species	Appearance
MRa404	B	1.3 g	TB1168 (pilot plant #1)	1.9 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	37 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.46 g 36%	13.6 ml 29.6 ml/g	Mostly 2	shiny black/grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa405	A	1.0 g	TB1168 (pilot plant #1)	1.2 g KOH (90%) 0.31 g NaOH (98%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.23 g 22%	14.0 ml 61.6 ml/g	2	shiny black/grey very voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MB3056	C	55.0 g	TB1168 (pilot plant #1) + H2O	82.6 g KOH (90%)	blender, 10 + 3x30 sec	yes	N2	3 °C/min	800 °C	1 h	600 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	26.5 g 48%	145 ml 5.5 ml/g	1	dull black/grey
		Rotating oven, powder not compacted														
MB3057	C	51.9 g	TB1168 (pilot plant #1) + H2O	77.8 g KOH (90%)	blender, 10 + 3x30 sec	yes	N2	3 °C/min	800 °C	1 h	547 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	25.2 g 49%	155 ml 6.2 ml/g	1	dull black/grey
		Rotating oven, powder not compacted														
MRa406	A	1.3 g	TB1168 (pilot plant #1)	1.9 g KOH (90%)	mill (3 min)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.39 g 31%	30.0 ml 76.5 ml/g	2	shiny black/grey very voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa407	B	1.0 g	TB1168 (pilot plant #1) + H2O	1.6 g KOH (90%)	mill (3 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.43 g 41%	13.1 ml 30.5 ml/g	1/2 mix	shiny black/grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa408	C	28.4 g	TB1168 (pilot plant #1)	34.0 g KOH (90%) 8.5 g NaOH (98%)	blender, 3x30 sec	no	N2	3 °C/min	800 °C	1 h	471 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	14.1 g 50%	320 ml 22.7 ml/g	2	shiny black/grey voluminous
		Rotating oven, powder not compacted														
MB3058	C	51.1 g	TB1168 (pilot plant #1) + H2O	76.7 g KOH (90%)	blender, 10 + 3x30 sec	yes	N2	3 °C/min	800 °C	1 h	548 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	24.4 g 48%	135.0 ml 5.5 ml/g	1	dull black/grey
		Rotating oven, powder not compacted														
MRa409A	B	1.1 g	TB1168 (pilot plant #1)	1.7 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	42 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.44 g 40%	2.8 ml 6.3 ml/g	Mostly 1	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa409B	A	0.8 g	TB1168 (pilot plant #1)	1.3 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	32 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.35 g 42%	8.2 ml 23.2 ml/g	1/2 mix	shiny black/grey voluminous
		ground biochar (brown powder), NOT compacted														
MRa410	C	48.4 g	TB1168 (pilot plant #1) + H2O	72.7 g KOH (90%)	blender, 10 + 3x30 sec	yes	N2	3 °C/min	800 °C	1 h	502 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	24.0 g 50%	110.0 ml 4.6 ml/g	1	dull black/grey
		Rotating oven, powder not compacted														
MRa411A	A	1.1 g	TB1170 (pilot plant #2)	1.7 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	42 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.43 g 38%	12.0 ml 28.0 ml/g	mostly 2	shiny black/grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa411B	B	0.9 g	TB1170 (pilot plant #2)	1.3 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	33 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.35 g 40%	6.4 ml 18.2 ml/g	1/2 mix	shiny black/grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size														
MRa412	C	49.2 g	TB1168 (pilot plant #1) + H2O	73.7 g KOH (90%)	blender, 10 + 3x30 sec	no	N2	3 °C/min	800 °C	1 h	553 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	23.9 g 49%	110.0 ml 4.6 ml/g	1	dull black/grey
		Rotating oven, powder not compacted														
MB3060	C	46.4 g	TB1168 (pilot plant #1) + H2O	69.6 g KOH (90%)	blender, 10 + 3x30 sec	yes	N2	3 °C/min	800 °C	1 h	458 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	22.6 g 49%	120.0 ml 5.3 ml/g	1	dull black/grey
		Rotating oven, powder not compacted														

Experiment	Oven	Qty	Lot	Base	grinding	homogeneous?	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Vol/Dens	Species	Appearance
MB413A	A	1.0 g	TB1168 (pilot plant #1)	1.4 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	21 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.40 g 41%	14.2 ml 35.7 ml/g	1/2 mix	shiny black/grey voluminous
			ground biochar (brown powder) compacted to two pellets of equal size													
MB413B	B	1.0 g	TB1168 (pilot plant #1)	1.4 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	21 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.41 g 43%	7.2 ml 17.5 ml/g	1/2 mix	shiny black/grey voluminous
			ground biochar (brown powder) compacted to two pellets of equal size													
MB414A	A	1.0 g	TB1168 (pilot plant #1)	1.5 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	13 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.40 g 40%	2.1 ml 5.3 ml/g	1	dull black/grey
			ground biochar (brown powder) compacted to two pellets of equal size													
MB414B	B	0.5 g	TB1168 (pilot plant #1)	0.8 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	9 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.24 g 46%	0.9 ml 3.8 ml/g	1	dull black/grey
			ground biochar (brown powder) compacted to one pellet													
MB3062	C	26.8 g	TB1168 (pilot plant #1) 1.4% H2O	32.2 g KOH (90%) 8.0 g NaOH (98%)	blender, 3x30 sec	no	N2	3 °C/min	800 °C	1 h	357 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	13.7 g 51%	260 ml 19.0 ml/g	2	shiny black/grey voluminous
			Rotating oven, powder not compacted													
MR415	C	28.8 g	TB1168 (pilot plant #1)	34.6 g KOH (90%) 8.7 g NaOH (98%)	blender, 3x30 sec	no	N2	3 °C/min	800 °C	1 h	431 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	14.4 g 50%	260 ml 18.1 ml/g	2	shiny black/grey voluminous
			Rotating oven, powder not compacted													
MB416A	A	0.5 g	(various combined lots)	0.7 g KOH (90%)	mill (3 min)	no	N2	3 °C/min	800 °C	1 h	7 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.23 g 49%	1.0 ml 4.3 ml/g	1	dull black/grey
			ground biochar (brown powder) compacted to two pellets of equal size													
MB416B	B	0.5 g	(various combined lots)	0.7 g KOH (90%)	mill (3 min)	no	N2	3 °C/min	800 °C	1 h	8 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.22 g 46%	1.0 ml 4.5 ml/g	1	dull black/grey
			ground biochar (brown powder) compacted to two pellets of equal size													
MB3063	C	47.8 g	TB1168 (pilot plant #1) + H2O	71.6 g KOH (90%)	blender, 10 + 3x30 sec	yes	N2	3 °C/min	800 °C	1 h	466 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	22.8 g 48%	130.0 ml 5.7 ml/g	1	dull black/grey
			Rotating oven, powder not compacted													
MB417A	A	0.5 g	(various combined lots)	0.7 g KOH (90%)	mill (3 min)	no	N2	3 °C/min	800 °C	1 h	8 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.19 g 39%	1.0 ml 5.3 ml/g	1	dull black/grey
			ground biochar (brown powder) compacted to two pellets of equal size													
MB417B	B	0.5 g	(various combined lots)	0.7 g KOH (90%)	mill (3 min)	no	N2	3 °C/min	800 °C	1 h	8 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.20 g 42%	1.7 ml 8.5 ml/g	Mostly 1	dull black/grey
			ground biochar (brown powder) compacted to two pellets of equal size													
MB3064	C	47.2 g	TB1168/1170 (pilot plant #1) + H2O	70.8 g KOH (90%)	blender, 10 + 3x30 sec	yes	N2	3 °C/min	800 °C	1 h	493 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	21.4 g 45%	120.0 ml 5.6 ml/g	1	dull black/grey
			Rotating oven, powder not compacted													
MB3065	C	49.0 g	TB1170 (pilot plant #2) + H2O	73.4 g KOH (90%)	blender, 10 + 3x30 sec	no	N2	3 °C/min	800 °C	1 h	514 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	22.3 g 46%	120.0 ml 5.4 ml/g	1	dull black/grey
			Rotating oven, powder not compacted, drinking water used in workup													
MB418A	A	0.5 g	(various combined lots)	0.7 g KOH (90%)	mill (3 min)	no	N2	3 °C/min	800 °C	1 h	8 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.17 g 35%	1.0 ml 6.0 ml/g	1	dull black/grey
			ground biochar (brown powder) compacted to two pellets of equal size													
MB418B	B	0.5 g	(various combined lots)	0.7 g KOH (90%)	mill (3 min)	no	N2	3 °C/min	800 °C	1 h	7 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.18 g 38%	1.0 ml 5.5 ml/g	1	dull black/grey
			ground biochar (brown powder) compacted to two pellets of equal size													

Experiment	Oven	Qty	Lot	Base	grinding	homogeneous?	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Vol/Dens	Species	Appearance
MB3066	C	49.2 g	TB1170 (pilot plant #2) + H2O	73.9 g KOH (90%)	blender, 10 + 3x30 sec	no	N2	3 °C/min	800 °C	1 h	479 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	23.8 g 48%	130.0 ml 5.5 ml/g	1	dull black/grey
			Rotating oven, powder not compacted													
MB419A	A	0.4 g	TB1170 (pilot plant #2) + H2O	0.7 g KOH (90%)	mill (1+3 min)	no	N2	3 °C/min	800 °C	1 h	8 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.11 g 26%	2.7 ml 24.0 ml/g	2	shiny black/grey voluminous
			ground biochar (brown powder) compacted to two pellets of equal size													
MB419B	B	0.5 g	TB1170 (pilot plant #2) + H2O	0.7 g KOH (90%)	mill (1+3 min)	no	N2	3 °C/min	800 °C	1 h	8 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.12 g 24%	2.7 ml 23.3 ml/g	Mostly 2	shiny black/grey voluminous
			ground biochar (brown powder) compacted to two pellets of equal size													
MRa420	C	49.3 g	TB1170 (pilot plant #2) + H2O	73.9 g KOH (90%)	blender, 10 + 3x30 sec	no	N2	3 °C/min	800 °C	1 h	588 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	23.3 g 47%	135.0 ml 5.8 ml/g	1	dull black/grey
			Rotating oven, powder not compacted													
MB421A	A	0.4 g	TB1170 (pilot plant #2) + H2O	0.7 g KOH (90%)	mill (1+3 min)	no	N2	3 °C/min	800 °C	1 h	8 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.10 g 23%	2.8 ml 27.2 ml/g	Mostly 2	shiny black/grey voluminous
			ground biochar (brown powder) compacted to two pellets of equal size													
MB421B	B	0.4 g	TB1170 (pilot plant #2) + H2O	0.7 g KOH (90%)	mill (1+3 min)	no	N2	3 °C/min	800 °C	1 h	7 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.11 g 26%	3.2 ml 28.1 ml/g	1/2 mix	shiny black/grey voluminous
			ground biochar (brown powder) compacted to two pellets of equal size													
MRa423	C	49.4 g	TB1170 (pilot plant #2) + H2O	74.1 g KOH (90%)	blender, 10 + 3x30 sec	yes	N2	3 °C/min	800 °C	1 h	555 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	23.5 g 48%	130.0 ml 5.5 ml/g		dull black/grey
			Rotating oven, powder not compacted													
MB3067	C	46.6 g	TB1170 (pilot plant #2) + H2O	70.0 g KOH (90%)	blender, 10 + 3x30 sec	yes	N2	3 °C/min	800 °C	1 h	468 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	21.93 g 47%	125.0 ml 5.7 ml/g		dull black/grey
			Rotating oven, powder not compacted													
MB3068	C	45.6 g	TB1170 (pilot plant #2) + H2O	68.4 g KOH (90%)	blender, 10 + 3x30 sec	yes	N2	3 °C/min	800 °C	1 h	454 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	21.4 g 47%	120.0 ml 5.6 ml/g		dull black/grey
			Rotating oven, powder not compacted													

Comments

- Yellow/red text for the lot of biochar show cases where the quality of the milled material was somewhat/significantly impaired by contact with air and/or moisture (leading to heating or colour change of the material, stickiness, clumping or similar changed in appearance)
- The information on homogeneity of the milled material is based on optical and acoustic evaluation